

Hands on Lab: Use the Windows Command Prompt for Administration



Estimated time needed: 45 minutes

About This Lab

In this lab we will use Microsoft Windows command prompt tools for various administrative tasks.

Objectives

In this hands-on lab, you will:

- Obtain system information using the command line.
- Manage directories using the command line.
- Perform system maintenance using the command line.

Important Information About Lab Instructions and Solutions

In case you try to use your physical keyboard in the lab environment, it might not produce any visible results. To avoid this issue, please use the On-Screen Keyboard (you can find it by searching for On-Screen Keyboard in the search bar at the bottom of your screen). If search functionality doesn't work, you can also click on the Windows icon, scroll down to find Windows Ease of Access, click on it, and then select On-Screen Keyboard.

Important Notices about This Lab

About Lab Sessions

Lab sessions are not persisted. This means that every time you connect to this lab, a new environment is created for you. Any data or files you saved in a previous session are no longer available. To avoid losing your data, plan to complete these tasks in a single session.

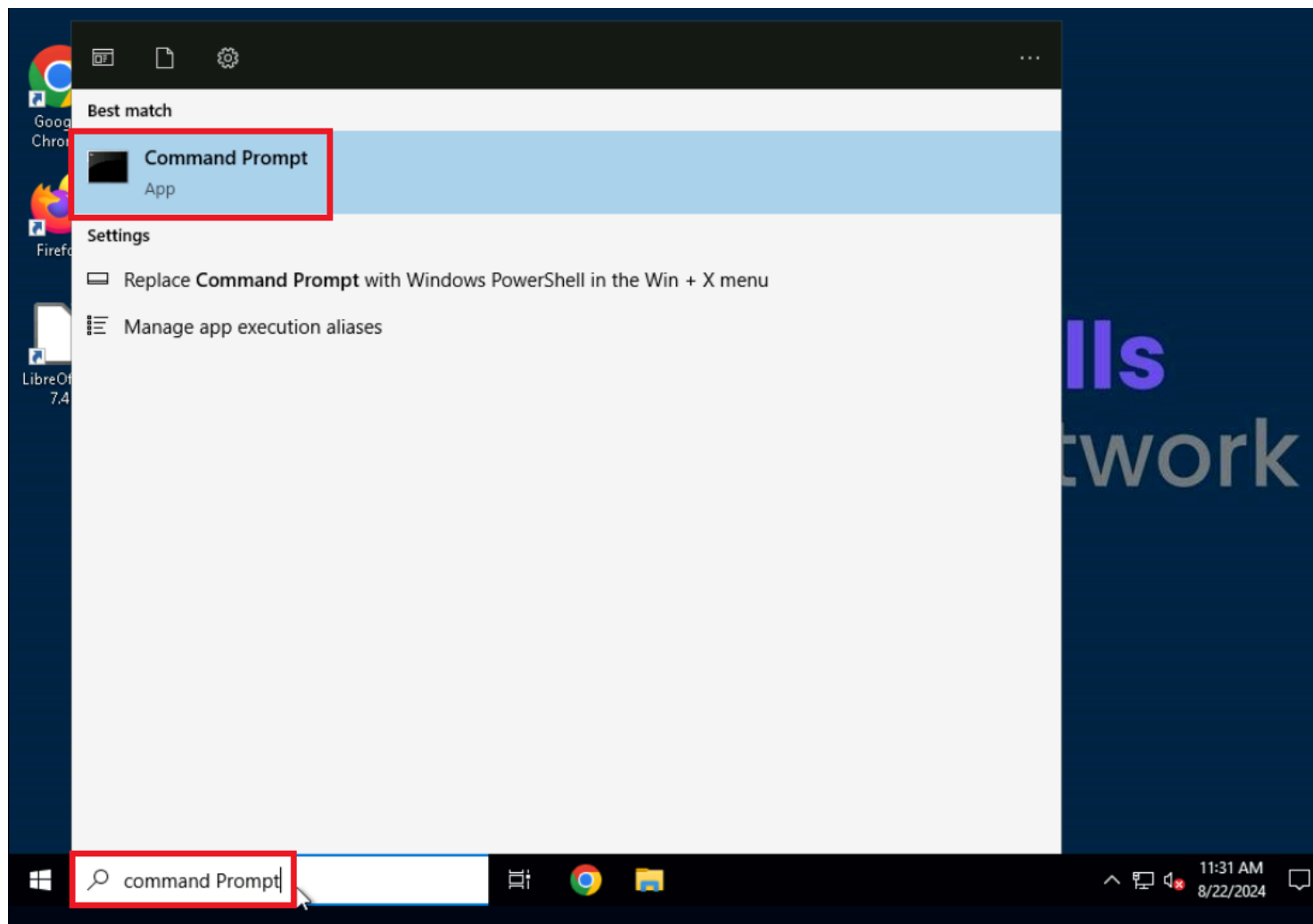
About the Lab Instructions and Solutions

Microsoft Windows operating system features can vary based on the Windows edition. If completing these exercises on your machine, your navigation and solutions may differ from what's presented in this lab.

Exercise 1: Obtain system information using the command line

In this exercise we will obtain information about the system and network using command line tools.

1. Type **Command Prompt** in the search bar and click on it.



2. Type **hostname** into the command line prompt and press **Enter**.

```
C:\> Administrator: Command Prompt

Microsoft Windows [Version 10.0.17763.1457]
(c) 2018 Microsoft Corporation. All rights reserved.

C:\Users\Administrator>hostname
EC2AMAZ-J5GP1U8

C:\Users\Administrator>
```

3. Type **ipconfig** into the command line prompt and press **Enter**.

```
Administrator: Command Prompt

Microsoft Windows [Version 10.0.17763.1457]
(c) 2018 Microsoft Corporation. All rights reserved.

C:\Users\Administrator>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet 3:

    Connection-specific DNS Suffix  . : ec2.internal
    Link-local IPv6 Address . . . . . : fe80::40a6:d85d:3058:fecf%5
    IPv4 Address. . . . . : 10.199.48.169
    Subnet Mask . . . . . : 255.255.128.0
    Default Gateway . . . . . : 10.199.0.1
```

4. Here you will see several network identifications for the local system including IP address, subnet mask, and default gateway. Type **ipconfig /all** into the command prompt to see more detailed network information.

```
C:\Users\Administrator>ipconfig/all

Windows IP Configuration

    Host Name . . . . . : EC2AMAZ-J5GP1U8
    Primary Dns Suffix . . . . . :
    Node Type . . . . . : Hybrid
    IP Routing Enabled. . . . . : No
    WINS Proxy Enabled. . . . . : No
    DNS Suffix Search List. . . . . : us-east-1.ec2-utilities.amazonaws.com
                                     eu-west-1.ec2-utilities.amazonaws.com
                                     eu-west-1.compute.internal
                                     ec2.internal

Ethernet adapter Ethernet 3:

    Connection-specific DNS Suffix  . : ec2.internal
    Description . . . . . : Amazon Elastic Network Adapter
    Physical Address. . . . . : 0A-1B-37-89-60-E3
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes
    Link-local IPv6 Address . . . . . : fe80::40a6:d85d:3058:fecf%5(Preferred)
    IPv4 Address. . . . . : 10.199.48.169(Preferred)
    Subnet Mask . . . . . : 255.255.128.0
    Lease Obtained. . . . . : Thursday, January 18, 2024 7:41:01 AM
    Lease Expires . . . . . : Thursday, January 18, 2024 8:41:02 AM
    Default Gateway . . . . . : 10.199.0.1
    DHCP Server . . . . . : 10.199.0.1
    DHCPv6 IAID . . . . . : 134405690
    DHCPv6 Client DUID. . . . . : 00-01-00-01-26-20-AB-4F-0E-9A-11-9F-E5-77
    DNS Servers . . . . . : 10.199.0.2
    NetBIOS over Tcpip. . . . . : Enabled
```

5. Type **cls** to clear the command line window.

```
Ethernet adapter Ethernet 3:

Connection-specific DNS Suffix  . : ec2.internal
Description . . . . . : Amazon Elastic Network Adapter
Physical Address. . . . . : 0A-1B-37-89-60-E3
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::40a6:d85d:3058:fecf%5(Preferred)
IPv4 Address. . . . . : 10.199.48.169(Preferred)
Subnet Mask . . . . . : 255.255.128.0
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DNS Servers . . . . . : 10.199.0.2
NetBIOS over Tcpip. . . . . : Enabled

C:\Users\Administrator>cls
```

Exercise 2: Manage directories using the command line

In this exercise we will view directories, create directories, and remove directories using the command line.

1. Type **dir** into the command line prompt and press **Enter**.

```
C:\Users\Administrator>dir
Volume in drive C has no label.
Volume Serial Number is F455-D01B

Directory of C:\Users\Administrator

09/16/2020  11:45 AM    <DIR>          .
09/16/2020  11:45 AM    <DIR>          ..
04/10/2020  03:43 PM             126 .gitconfig
04/13/2020  05:03 PM    <DIR>          .ssh
04/10/2020  02:57 PM    <DIR>          .vscode
09/16/2020  11:52 AM    <DIR>          3D Objects
09/16/2020  11:52 AM    <DIR>          Contacts
09/16/2020  11:52 AM    <DIR>          Desktop
09/16/2020  11:52 AM    <DIR>          Documents
01/17/2024  11:56 AM    <DIR>          Downloads
09/16/2020  11:52 AM    <DIR>          Favorites
09/16/2020  11:52 AM    <DIR>          Links
09/16/2020  11:52 AM    <DIR>          Music
09/16/2020  11:52 AM    <DIR>          Pictures
09/16/2020  11:52 AM    <DIR>          Saved Games
09/16/2020  11:52 AM    <DIR>          Searches
01/17/2024  10:05 AM    <DIR>          Videos
04/10/2020  02:56 PM    <DIR>          vimfiles
04/28/2020  08:38 PM             13,957 _viminfo
               2 File(s)              14,083 bytes
              17 Dir(s)  27,924,746,240 bytes free
```

2. Here you can see the contents of the current directory. To change to the **Videos** directory type **cd videos**.

```
C:\Users\Administrator>dir
Volume in drive C has no label.
Volume Serial Number is F455-D01B

Directory of C:\Users\Administrator

09/16/2020  11:45 AM    <DIR>          .
09/16/2020  11:45 AM    <DIR>          ..
04/10/2020  03:43 PM      126 .gitconfig
04/13/2020  05:03 PM    <DIR>          .ssh
04/10/2020  02:57 PM    <DIR>          .vscode
09/16/2020  11:52 AM    <DIR>          3D Objects
09/16/2020  11:52 AM    <DIR>          Contacts
09/16/2020  11:52 AM    <DIR>          Desktop
09/16/2020  11:52 AM    <DIR>          Documents
01/17/2024  11:56 AM    <DIR>          Downloads
09/16/2020  11:52 AM    <DIR>          Favorites
09/16/2020  11:52 AM    <DIR>          Links
09/16/2020  11:52 AM    <DIR>          Music
09/16/2020  11:52 AM    <DIR>          Pictures
09/16/2020  11:52 AM    <DIR>          Saved Games
09/16/2020  11:52 AM    <DIR>          Searches
01/17/2024  10:05 AM    <DIR>          Videos
04/10/2020  02:56 PM    <DIR>          vimfiles
04/28/2020  08:38 PM    13,957 _viminfo
                2 File(s)          14,083 bytes
                17 Dir(s)  27,924,746,240 bytes free

C:\Users\Administrator>cd videos
```

3. Note that the command prompt has changed indicating that you are now in the **C:/Users/Administrator/Videos** directory. To create a new directory, type **mkdir comedy** into the command line prompt and press **Enter**.

```
C:\Users\Administrator\Videos>mkdir comedy

C:\Users\Administrator\Videos>dir
Volume in drive C has no label.
Volume Serial Number is F455-D01B
```

4. To view your newly created directory, type **dir** and press **Enter**.

```
C:\Users\Administrator\Videos>dir
Volume in drive C has no label.
Volume Serial Number is F455-D01B

Directory of C:\Users\Administrator\Videos

01/18/2024  08:00 AM    <DIR>          .
01/18/2024  08:00 AM    <DIR>          ..
01/18/2024  08:00 AM    <DIR>          comedy
                0 File(s)          0 bytes
                3 Dir(s)  27,926,732,800 bytes free
```

5. To delete the newly created directory, type **rmdir comedy** and press **Enter**.

```
C:\Users\Administrator\Videos>rmdir comedy  
C:\Users\Administrator\Videos>
```

6. To verify that the directory has been removed, type **dir** and press **Enter**.

```
C:\Users\Administrator\Videos>rmdir comedy  
C:\Users\Administrator\Videos>dir  
Volume in drive C has no label.  
Volume Serial Number is F455-D01B  
  
Directory of C:\Users\Administrator\Videos  
  
01/18/2024  08:04 AM    <DIR>          .  
01/18/2024  08:04 AM    <DIR>          ..  
                0 File(s)                0 bytes  
                2 Dir(s)  28,036,890,624 bytes free  
C:\Users\Administrator\Videos>
```

7. To return to the previous directory, type **cd ..**

```
C:\Users\Administrator\Videos>cd ..  
C:\Users\Administrator>
```

Exercise 3: Perform system maintenance using the command line

In this exercise we will check the file system for errors and shut down the local system using the command line.

1. Type **chkdsk** into the command line prompt. This process will take several minutes as chkdsk checks the file systems for logical and physical errors.

```
C:\Users\Administrator>chkdsk  
The type of the file system is NTFS.  
  
WARNING!  /F parameter not specified.  
Running CHKDSK in read-only mode.  
  
Stage 1: Examining basic file system structure ...  
  413440 file records processed.  
File verification completed.  
  6072 large file records processed.  
  0 bad file records processed.  
  
Stage 2: Examining file name linkage ...  
  112 reparse records processed.  
  602888 index entries processed.  
Index verification completed.
```

Once complete, you can review the results of the chkdsk. As indicated, no errors were found, and no further action is required.

Congratulations!

You have completed this lab and are ready for the next topic.

Authors

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